

EXPERIMENT-15

SWAPING OF TWO 8-BIT DATA.

Aim: To write an assembly language program to swap two 8-bit data using 8085 processor.

Algorithm:

1. load first data into A and B.
2. load second data into A and C.
3. Store second data at 1102H.
4. move B to A.
5. store first data at 1103H.
6. Stop program.

PROGRAM:

LDA 1100 - load first data.
MOV B, A - Save data in B
LDA 1101 - load second data.
MOV C, A - Save data in C.
STA 1102 - store second data.
MOV A, B - restore first data.
STA 1103 - store first data.
HLT - stop program.

Input:

Address	data.
1100	6
1101	4

Output:

Address	data.
1103	4
1104	6

Result: Thus the program was executed successfully using 8085 processor simulator

GNUSim8085 - 8085 Microprocessor Simulator

FileResetAssemblerDebugHelp

Registers

A06

BC0604

DE0000

HL0000

PSW0000

PC4210

SPFFFF

Int-Reg00

Flag

S0

Z0

AC0

P0

C0

Decimal - Hex Conversion

DecimalHex

00

To HexTo Dec

I/O Ports

0-+00

Update Port Value

Memory

1101-+04

Update Memory

Load me at

1LDA 1100

2MOV B, A

3LDA 1101

4MOV C, A

5STA 1102

6MOV A, B

7STA 1103

8HLT

DataStackKeyPadMemoryI/O Ports

Start1102OK

Address (Hex)AddressData

044E11024

044F11036

045011040

045111050

045211060

045311070

045411080

045511090

045611100

045711110

045811120

045911130

045A11140

045B11150

045C11160

045D11170

Line NoAssembler Message

0Program assembled successfully

Simulator: Idle